

Pengxiang (Eric) Wang

ericwang7717@gmail.com · linkedin.com/in/-ericwang- · github.com/W-Kaski · ek-flowity.site

EDUCATION

University of Toronto

Sep 2023 – Apr 2027 (Expected)

B.Sc. Computer Science (Specialist) GPA: 3.77 / 4.00

Coursework: Algorithm Design & Analysis, Operating Systems, Probability & Statistics, Databases, Deep Learning, Reinforcement Learning

TECHNICAL SKILLS

LLM Systems:	RAG pipelines, ReAct agent loop, tool calling, MCP protocol, SSE token streaming
Vector Search:	PgVector (HNSW, cosine), query rewriting, hybrid retrieval, DashScope embeddings
Inference Infra:	Spring AI, LangChain4j, qwen-plus / DashScope, Ollama (local inference)
Backend:	Java, Spring Boot, MyBatis Plus, PostgreSQL, Redis, Docker
Reliability:	Sa-Token RBAC, Guava token-bucket rate limiter, Caffeine cache, OWASP sanitizer
ML:	PyTorch, HuggingFace Transformers, model distillation, training pipelines
Frontend:	TypeScript, React, Vue

PROJECTS

EK AI Agent — Full-Stack LLM Agent Platform

Spring AI 1.0 · Java 21 · PgVector · SSE · MCP · Docker | [GitHub](#)

- Architected a dual-application LLM platform — domain RAG chatbot + autonomous ReAct agent with 7 pluggable tools — delivering **~60% reduction in time-to-first-token** via SSE streaming under a 20-step deterministic execution budget.
- Improved RAG retrieval precision by **~45%** through a two-stage pipeline: pre-retrieval LLM query rewriting + index-time keyword enrichment per chunk, enabling future hybrid vector-keyword search without re-indexing.
- Designed a 4-layer agent hierarchy (**BaseAgent** → **ReActAgent** → **ToolCallAgent** → **EkManus**); disabled Spring AI's internal tool execution to enforce deterministic loop control with explicit **TerminateTool**-based exit.
- Scaled concurrent agent session capacity **~2.5×** by routing ReAct loops through **CompletableFuture** async execution on Java 21's virtual-thread-compatible runtime, keeping SSE streams non-blocking.
- Extended tool surface to 4 external services (web search, scraping, file I/O, PDF generation) + MCP sidecar over stdio/SSE; selected PgVector (HNSW) over in-memory store for persistent, query-able vectors with only +10ms retrieval latency trade-off.

Cloud Gallery — Collaborative Cloud Image Platform

Spring Boot 2.7 · Java 11 · Vue 3 · WebSocket · Disruptor · Tencent COS · Redis | [GitHub](#)

- Eliminated cross-tenant data leakage across concurrent sessions by isolating two independent Sa-Token **StpLogic** instances (system roles vs. space roles), enforced at every API boundary without coupling the two domains.
- Achieved deterministic conflict resolution for real-time collaborative edits by routing concurrent WebSocket writes through a Disruptor-backed event queue with a lock-editor protocol.
- Reduced image post-processing compute cost by configuring Tencent COS CI rules to auto-trigger WebP compression (~30% size reduction), thumbnail generation, and dominant-color extraction on every upload.
- Delivered color-based image search at **single-query latency**, no external embedding model, by normalizing dominant colors to HSL and ranking via Euclidean RGB distance in a SQL aggregation.

What2Do — Campus Activity Platform

Spring Boot 3.2 · Java 17 · React 19 · PostgreSQL · Caffeine · Cloudflare R2

- Enforced 3-tier RBAC (USER / CLUB_MANAGER / ADMIN) across 40+ endpoints via Sa-Token AOP + React Router **ProtectedRoute**, closing client-side auth bypass at both the API and UI layers.
- Cut repeated DB reads for geo/event data by **~80%** with 3 independently tuned Caffeine **CacheManager** beans (TTLs: 24h / 10min / 15min); cache hit rates surfaced via Spring Actuator.
- Hardened the API with a 5-layer security chain (IP blacklist → rate limiting [per-IP / per-user / per-endpoint] → auth → AOP role → OWASP XSS sanitizer).

NeuTTS-Express — Expressive TTS via Synthetic Distillation

PyTorch · Knowledge Distillation · HuggingFace Transformers · Neural Audio Codec · NeurIPS Course Project | [GitHub](#)

- Built the evaluation pipeline (Whisper WER, RTF, batch stability) for a 0.5B student TTS model distilled from a 3B teacher; student ran **2.16×** **faster than real-time** with **~4×** **lower WER variance** (std 2.10% vs. 7.91%) than the larger baseline.
- Conducted an 11-participant MOS listening study across 7 emotions; student outperformed the 6× larger teacher on naturalness (3.99 vs. 3.17) and emotion match (3.91 vs. 2.82).